

Optimal Mixing on Random Regular Graphs beyond Uniqueness

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Abstract

Building on the spectral independence established by Chen et al. [1], we strengthen the corresponding Poincaré inequality to approximate tensorization of entropy (ATE) for the hard-core model on random Δ -regular graphs beyond the tree-uniqueness threshold. For fixed $\Delta \geq 3$ and $\delta \in (0, 1)$, the ATE constant is at most $\exp(O_\delta(\sqrt{\Delta}))$ with high probability when $\lambda \leq (1 - \delta)/(2\sqrt{\Delta} - 1 - 1)$. The proof combines existing product domination, negative-fields localization, and ATE transfer arguments. We compute an explicit ATE bound for the easy low-fugacity measure, uniformly over pinnings, and integrate the entropy loss along the negative-field localization path to transfer this bound to the original measure. Consequently, Glauber dynamics mixes from every initial configuration in $e^{O_\delta(\sqrt{\Delta})}n(\log(n) + \log \log \frac{1}{\lambda})$ steps.

1 Introduction

The hard-core model is one of the canonical models at the interface of statistical physics, probability, combinatorics, and theoretical computer science. Given a graph $G = (V, E)$ and a fugacity $\lambda > 0$, the hard-core distribution $\mu_{G,\lambda}$ is the Gibbs measure on the family $\mathcal{I}(G)$ of independent sets of G , defined by

$$\mu_{G,\lambda}(I) = \frac{\lambda^{|I|}}{Z_G(\lambda)}, \quad Z_G(\lambda) = \sum_{J \in \mathcal{I}(G)} \lambda^{|J|}.$$

For graphs of maximum degree $\Delta \geq 3$, the worst-case behavior is governed by the tree-uniqueness threshold $\lambda_c(\Delta) = (\Delta - 1)^{\Delta-1}/(\Delta - 2)^\Delta \sim e/\Delta$. Below this threshold, Glauber dynamics mixes in $O(n \log n)$ steps with a fixed gap to uniqueness [2, 3, 4, 5, 6].

Despite the worst-case hardness beyond the uniqueness threshold [7, 8], polynomial-time sampling is expected for many other graph families, most notably random Δ -regular graphs. Using a trickle-down argument, [1] derived spectral independence from spectral properties of the adjacency matrix of a random Δ -regular graph and obtained an $O_\delta(n^2 \log \Delta)$ mixing-time bound. More recently, [9] gave a short, self-contained proof of the corresponding Poincaré inequality via a Bochner–Bakry–Émery argument. Also, [10, 11] recently developed efficient sampling algorithms for the hard-core model on random regular bipartite graphs beyond the uniqueness threshold.

In this paper, we establish a stronger functional inequality, approximate tensorization of entropy, for the hard-core model on random Δ -regular graphs beyond the uniqueness threshold, and obtain an $e^{O_\delta(\sqrt{\Delta})}n \log(en)$ mixing-time bound for Glauber dynamics from arbitrary initial configurations. Our argument builds on the spectral independence property established in [1], the negative-fields localization framework of [6], and an approximate tensorization of entropy (ATE) transfer argument from [12].

Write $C_{\text{AT}}(\mu)$ for the ATE constant and $\alpha_{\text{Ent}}(P_\mu^{\text{GD}})$ for the one-step entropy-contraction coefficient, as defined in Section 2. Throughout, T_{mix} denotes the worst-case total-variation mixing time at distance $1/4$ for discrete-time random-scan Glauber dynamics.

Theorem 1.1 (Hard-core ATE and arbitrary-start mixing). *Fix $\delta \in (0, 1)$ and $\Delta \geq 3$. Let G be an n -vertex graph of maximum degree Δ with $a = -\lambda_{\min}(A_G) > 1$. Set $x = \Delta/(a - 1)$. If*

$$\lambda(a - 1) \leq 1 - \delta,$$

then

$$C_{\text{AT}}(\mu_{G,\lambda}) \leq \exp(O_\delta(x)). \quad (1)$$

Consequently,

$$\alpha_{\text{Ent}}(P_\mu^{\text{GD}}) \geq \frac{1}{n} \exp(-O_\delta(x)),$$

and the Glauber dynamics satisfies

$$T_{\text{mix}} \leq \exp(O_\delta(x)) n(\log n + \log \log \frac{1}{\lambda}). \quad (2)$$

If $a = \Theta(\sqrt{\Delta})$, the prefactor in (2) is $\exp(O_\delta(\sqrt{\Delta}))$.

Corollary 1.2 (Random regular graphs, fixed degree). *Fix an integer $\Delta \geq 3$ and $\delta \in (0, 1)$, and let $n \rightarrow \infty$ through admissible values. Let G be a uniformly random Δ -regular graph. If*

$$\lambda \leq \frac{1 - \delta}{2\sqrt{\Delta - 1} - 1},$$

then, with probability $1 - o_n(1)$,

$$T_{\text{mix}} \leq e^{O_\delta(\sqrt{\Delta})} n(\log n + \log \log \frac{1}{\lambda}).$$

The key estimate is an integrated entropy cost of $O_\delta(\Delta\lambda)$, rather than the cost obtained by freezing the marginal-stability bound at the initial field. We prove the transfer statement in [Section 2](#) and its hard-core application in [Section 3](#); the easy-regime ATE proof is given in [Section A](#).

2 Approximate tensorization via negative fields

2.1 Notation and the complete EI input

A binary configuration is identified interchangeably with $x \in \{0, 1\}^n$ and with the set $\{i : x_i = 1\}$. For a probability measure μ on a finite space and $f \geq 0$, write

$$\text{Ent}_\mu(f) = \mu[f \log f] - \mu(f) \log \mu(f), \quad 0 \log 0 = 0.$$

If $f = d\nu/d\mu$ and ν is a probability measure, then $\text{Ent}_\mu(f) = D_{\text{KL}}(\nu\|\mu)$. For a feasible partial configuration τ , μ^τ denotes the conditional measure on the unpinned coordinates; deterministic coordinates are regarded as pinned and removed. For $\xi \in \mathbb{R}_{>0}^n$, define

$$(\xi * \mu)(x) = \frac{\mu(x) \prod_i \xi_i^{x_i}}{\sum_y \mu(y) \prod_i \xi_i^{y_i}}, \quad \theta * \mu = (\theta \mathbf{1}) * \mu.$$

Write $p_i(\mu) = \mu(X_i = 1)$ and $R_i(\mu) = p_i(\mu)/(1 - p_i(\mu))$ on nondegenerate coordinates.

Let $K_i^\mu f = \mathbb{E}_\mu[f \mid X_{[n] \setminus \{i\}}]$ and $P_\mu^{\text{GD}} = \frac{1}{n} \sum_i K_i^\mu$. The single-site conditional entropy is $\text{Ent}_i(f) = K_i^\mu(f \log f) - K_i^\mu f \log K_i^\mu f$.

Definition 2.1 (Approximate tensorization of entropy). *The approximate tensorization entropy (ATE) constant $C_{\text{AT}}(\mu)$ is defined by the least C such that*

$$\text{Ent}_\mu(f) \leq C \sum_i \mu[\text{Ent}_i(f)] \quad \text{for every } f \geq 0. \quad (3)$$

Since $\mu[\text{Ent}_i(f)] = \text{Ent}_\mu(f) - \text{Ent}_\mu(K_i^\mu f)$, entropy convexity gives

$$\text{Ent}_\mu(P_\mu^{\text{GD}} f) \leq \left(1 - \frac{1}{nC_{\text{AT}}(\mu)}\right) \text{Ent}_\mu(f). \quad (4)$$

Thus the one-step entropy-contraction coefficient $\alpha_{\text{Ent}}(P_\mu^{\text{GD}})$, the largest coefficient α for which $\text{Ent}_\mu(P_\mu^{\text{GD}} f) \leq (1 - \alpha) \text{Ent}_\mu(f)$ for every $f \geq 0$, is at least $1/(nC_{\text{AT}}(\mu))$.

Definition 2.2 (Complete spectral independence and marginal stability [13, 5]). *Let $D_\pi = \text{diag}(p_i(\pi))$ on the nondegenerate coordinates. A binary measure μ is completely (η, ϵ) -spectrally independent if, for every feasible pinning τ and every $\xi \in (0, 1 + \epsilon]^{[n] \setminus \text{dom}(\tau)}$,*

$$\text{Cov}(\xi * \mu^\tau) \preceq \eta D_{\xi * \mu^\tau}. \quad (5)$$

A measure μ is ζ -marginally stable if, for every feasible pinning τ on Λ , every $U \subseteq \Lambda$, and every $i \notin \Lambda$ that is nondegenerate under μ^τ ,

$$R_i(\mu^\tau) \leq \zeta, \quad R_i(\mu^\tau) \leq \zeta R_i(\mu^{\tau_U}).$$

Let π and $\nu \ll \pi$ have one-site marginals p_i and q_i . We say that π is β -entropically independent if

$$\sum_i D_{\text{KL}}(\text{Ber}(q_i) \parallel \text{Ber}(p_i)) \leq \beta D_{\text{KL}}(\nu \parallel \pi) \quad \text{for every } \nu \ll \pi. \quad (6)$$

This is the down-to-one entropic-independence inequality after homogenization [5, 14].

Proposition 2.3 (Spectral independence to entropic independence [5]). *Let $\eta > 1$, $\epsilon > 0$, and $\zeta > 1$. Suppose π is completely (η, ϵ) -spectrally independent and ζ -marginally stable. Then every feasible pinning of π satisfies entropic independence for all absolutely continuous measures with*

$$\beta_{\text{F}} = \max \left\{ 2\eta, 1 + \frac{\log(2\zeta)}{\log(1 + \epsilon)} \right\}.$$

Remark 2.4. *This is the product-domination argument of [5, Lemmas 2.3 and 4.4]. The spectral independence defined in [5] uses a absolute influence matrix, while the proof relies on signed correlation matrix as in [13], which is the same as (5) here, see Lemma 5.8-5.10 in the proof of [5, Lemma 2.3].*

2.2 The transfer theorem

Theorem 2.5 (Full ATE transfer). *Let μ be a binary measure, let $\theta_0 \in (0, 1)$, and suppose that μ is completely (η, ϵ) -spectrally independent with $\eta > 1$ and $\epsilon > 0$. Assume that every feasible pinning of $\theta_0 * \mu$ satisfies ATE with constant C_0 , and that $\theta * \mu$ is $\zeta(\theta)$ -marginally stable for every $\theta \in [\theta_0, 1]$, where $\zeta : [\theta_0, 1] \rightarrow (1, \infty)$ is continuous. Set*

$$\beta_{\text{F}}(\theta) = \max \left\{ 2\eta, 1 + \frac{\log(2\zeta(\theta))}{\log(1 + \epsilon)} \right\}, \quad B_{\text{F}} = \int_{\theta_0}^1 \beta_{\text{F}}(\theta) \frac{d\theta}{\theta}. \quad (7)$$

Then $C_{\text{AT}}(\mu) \leq C_0 e^{B_{\text{F}}}$ and $\alpha_{\text{Ent}}(P_\mu^{\text{GD}}) \geq e^{-B_{\text{F}}} / (nC_0)$.

Proof. We use the negative-fields localization scheme of [6]. Sample $X \sim \mu$ and independent $U_i \sim \text{Unif}(0, 1)$ for occupied coordinates, and let $\tau_t = \{i \in X : U_i \leq t\}$. Let $\mathcal{F}_t$ be the observation filtration and $\mu_t = \mathcal{L}(X \mid \mathcal{F}_t)$. The posterior satisfies

$$\mu_t(x) \propto \mu(x) \mathbf{1}_{\{\tau_t \subseteq x\}} (1-t)^{|x \setminus \tau_t|},$$

so, after deleting the revealed occupied coordinates, $\mu_t = (1-t) * \mu^{\tau_t}$. In particular, (μ_t) is a measure-valued martingale. Set $T = 1 - \theta_0$.

Fix $f \geq 0$. Conditional on $\mathcal{F}_s$, write $\pi = \mu_s$ and $\xi = f\pi/\pi(f)$ when $\pi(f) > 0$; otherwise the conditional entropy vanishes. During $[s, s+h]$, every remaining occupied coordinate is revealed independently with probability $\varepsilon = h/(1-s)$. Let K_ε be this thinning kernel. The chain rule for relative entropy under the common observation kernel gives the exact identity

$$\text{Ent}_\pi(f) - \mathbb{E}[\text{Ent}_{\mu_{s+h}}(f) \mid \mathcal{F}_s] = \pi(f) D_{\text{KL}}(\xi K_\varepsilon \parallel \pi K_\varepsilon). \quad (8)$$

Indeed, let y be the revealed coordinates during $[s, s+h]$ and define π_y as the conditional distribution of π given observation y , then $\mu_{s+h} = \pi_y$, $y \sim \pi K_\varepsilon$ and

$$(\xi K_\varepsilon)(y) = \sum_x f(x) \frac{\pi(x) K_\varepsilon(x, y)}{\pi(f)} = \frac{(\pi K_\varepsilon)(y)}{\pi(f)} \sum_x f(x) \frac{\pi(x) K_\varepsilon(x, y)}{(\pi K_\varepsilon)(y)} = \frac{(\pi K_\varepsilon)(y) \pi_y(f)}{\pi(f)}.$$

Thus

$$\begin{aligned} \text{Ent}_{\pi K_\varepsilon}[\mathbb{E}_{\mu_{s+h}}[f] \mid \mathcal{F}_s] &= \sum_y (\pi K_\varepsilon)(y) \pi_y(f) \log \frac{\pi_y(f)}{\pi(f)} \\ &= \sum_y (\xi K_\varepsilon)(y) \pi(f) \log \frac{(\xi K_\varepsilon)(y)}{(\pi K_\varepsilon)(y)} \\ &= \pi(f) D_{\text{KL}}(\xi K_\varepsilon \parallel \pi K_\varepsilon). \end{aligned}$$

Write $p_i = \pi(X_i = 1)$ and $q_i = \xi(X_i = 1)$. For the empty and singleton observations,

$$(\pi K_\varepsilon)(\emptyset) = 1 - \varepsilon \sum_i p_i + O(\varepsilon^2), \quad (\pi K_\varepsilon)(\{i\}) = \varepsilon p_i + O(\varepsilon^2),$$

with the analogous formulas for ξ . Consequently,

$$D_{\text{KL}}(\xi K_\varepsilon \parallel \pi K_\varepsilon) = \varepsilon \sum_i \left(q_i \log \frac{q_i}{p_i} + p_i - q_i \right) + o(\varepsilon). \quad (9)$$

Each summand is bounded by $D_{\text{KL}}(\text{Ber}(q_i) \parallel \text{Ber}(p_i))$, because their difference is

$$(1 - q_i) \log \frac{1 - q_i}{1 - p_i} + q_i - p_i \geq 0.$$

The inequality follows from $a \log(a/b) + b - a \geq 0$ for $a \geq 0$, $b > 0$.

At external field $\theta = 1 - s$, complete spectral independence is inherited by the posterior: further fields compose to $\theta \xi \leq 1 + \epsilon$, and further pinnings are already included in the hypothesis. Marginal stability is likewise inherited by pinnings. Thus [proposition 2.3](#), (8), and (9) give

$$\mathbb{E}[\text{Ent}_{\mu_{s+h}}(f) \mid \mathcal{F}_s] \geq \left(1 - \frac{h\beta_F(1-s)}{1-s} \right) \text{Ent}_{\mu_s}(f) + o(h).$$

Furthermore, for $H(s) = \mathbb{E} \text{Ent}_{\mu_s}(f)$,

$$H'(s) \geq -\frac{\beta_F(1-s)}{1-s} H(s), \quad H(T) \geq e^{-B_F} \text{Ent}_{\mu}(f).$$

For a fixed f , the map $\pi \mapsto \pi[\text{Ent}_i^\pi(f)]$ is concave by [12, Lemma 29], applied to the observation that erases coordinate i . Since (μ_t) is a measure-valued martingale, Jensen's inequality gives $\mathbb{E} \mu_T[\text{Ent}_i^{\mu_T}(f)] \leq \mu[\text{Ent}_i^\mu(f)]$. Every terminal posterior is a pinning of $\theta_0 * \mu$, so the terminal ATE inequality yields

$$e^{-B_F} \text{Ent}_{\mu}(f) \leq \mathbb{E} \text{Ent}_{\mu_T}(f) \leq C_0 \sum_i \mathbb{E} \mu_T[\text{Ent}_i^{\mu_T}(f)] \leq C_0 \sum_i \mu[\text{Ent}_i^\mu(f)].$$

Pinned coordinates contribute zero. This proves the ATE bound, and (4) gives entropy contraction. $\square$

3 The hard-core model

We first give the marginal stability and spectral independence condition, and the terminal ATE inequality.

Lemma 3.1 (Hard-core marginal stability). *For the hard-core model at activity $\rho > 0$ on a graph of maximum degree $\Delta \geq 1$, every nondegenerate coordinate under every feasible pinning satisfies*

$$\frac{\rho}{(1+\rho)^\Delta} \leq R_v \leq \rho.$$

Consequently, the measure is $(1+\rho)^\Delta$ -marginally stable.

Proof. Every pinning leaves a hard-core model on an induced subgraph. The conditional odds at a remaining vertex satisfy

$$R_v = \rho \mathbb{P}(X_{N(v)} = 0 \mid X_v = 0).$$

The upper bound is immediate. Expose the neighbors sequentially and require each to be zero; every conditional probability is at least $1/(1+\rho)$. This gives the lower bound. $\square$

Lemma 3.2 (Complete spectral independence). *Let $a = -\lambda_{\min}(A_G) > 1$ and suppose $\lambda(a-1) \leq 1-\delta$ with $\delta \in (0,1)$. Then the hard-core measure is completely $(\frac{2}{\delta}, \frac{\delta}{2})$ -spectrally independent according to definition 2.2.*

Proof. Set $b = 1 + \frac{\delta}{2}$. Fix a feasible pinning and an external field $\xi \in (0, b]^V$. After deleting pinned and forced coordinates, the resulting measure $\tilde{\mu}$ is hard-core on an induced subgraph with activities $\lambda \xi_i$. For an independent set S , the active coordinates form another induced subgraph H . In the local dependency notation of [1, Theorem 1.14],

$$r_S(i) = \frac{\tilde{\mu}(S \cup \{i\})}{\tilde{\mu}(S)} = \lambda \xi_i, \quad M_S = -A_H,$$

where $M_S(i, i) = 0$ and, for $i \neq j$,

$$M_S(i, j) = \frac{\tilde{\mu}(S \cup \{i, j\}) \tilde{\mu}(S)}{\tilde{\mu}(S \cup \{i\}) \tilde{\mu}(S \cup \{j\})} - 1.$$

Interlacing gives $M_S \preceq aI$. Set $\delta' = 1 - b(1 - \delta) \geq \frac{\delta}{2}$. Since $\xi_i \leq b$ and $\lambda(a - 1) \leq 1 - \delta$,

$$M_S \preceq aI \preceq I + (1 - \delta') \operatorname{diag}(r_S)^{-1}.$$

The scalar trickle-down theorem [1, Theorem 1.14] therefore gives

$$\operatorname{Cov}(s * \tilde{\mu}) \preceq \frac{1}{1 - (1 - \delta')s} \operatorname{diag}(p_i(s * \tilde{\mu})), \quad 0 < s < 1.$$

Letting $s \uparrow 1$ yields $\operatorname{Cov}(\tilde{\mu}) \preceq (2/\delta) \operatorname{diag}(p_i(\tilde{\mu}))$. $\square$

Theorem 3.3 (Hard-core easy-regime ATE). *Let G have maximum degree $\Delta \geq 1$, and let μ_ρ be its hard-core measure at activity $0 < \rho \leq 1/\Delta$. Then, for every $f \geq 0$,*

$$\frac{1 - \rho(\Delta - 1)}{1 + \rho} \operatorname{Ent}_{\mu_\rho}(f) \leq \sum_{v \in V} \mu_\rho[\operatorname{Ent}_v(f)]. \quad (10)$$

In particular, at $\rho = 1/(2\Delta)$ every pinning satisfies ATE with constant

$$C_0 \leq \frac{1 + 1/(2\Delta)}{1 - (\Delta - 1)/(2\Delta)} = \frac{2\Delta + 1}{\Delta + 1} < 2.$$

The proof of [theorem 3.3](#) is given in [Section A](#); the full ATE inequality, not merely an MLSI, is needed at the terminal field.

Proof of Theorem 1.1. First, $a > 1$ implies $a \geq \sqrt{2}$. Indeed, a graph with no induced path on three vertices is a disjoint union of cliques and has minimum adjacency eigenvalue at least -1 . Thus G contains an induced P_3 , and interlacing gives $\lambda_{\min}(A_G) \leq -\sqrt{2}$. The fugacity condition therefore implies $\lambda \leq (1 - \delta)/(\sqrt{2} - 1) = O_\delta(1)$.

If $\lambda \leq 1/(2\Delta)$, the model is already in the easy regime and [theorem 3.3](#) gives $C_{\text{AT}} < 2$. Hence assume

$$\lambda > \frac{1}{2\Delta}, \quad \theta_0 = \frac{1}{2\Delta\lambda} < 1.$$

Every pinning of $\theta_0 * \mu$ is hard-core at activity $1/(2\Delta)$ on an induced subgraph, so [theorem 3.3](#) gives the uniform terminal constant $C_0 < 2$. By [lemma 3.2](#), the original measure is completely $(\frac{2}{\delta}, \frac{\delta}{2})$ -spectrally independent. By [lemma 3.1](#), the marginal-stability constant at scalar field θ is $\zeta(\theta) = (1 + \theta\lambda)^\Delta$. Thus the full-transfer exponent is

$$B_F = \int_{\theta_0}^1 \max \left\{ 2\eta, 1 + \frac{\log 2 + \Delta \log(1 + \theta\lambda)}{\log(1 + \frac{\delta}{2})} \right\} \frac{d\theta}{\theta}.$$

Using $\max\{u, v\} \leq u + v$ and $\log(1 + \theta\lambda) \leq \theta\lambda$ gives

$$\begin{aligned} B_F &\leq O_\delta(\log \theta_0^{-1}) + O_\delta \left(\Delta \int_{\theta_0}^1 \frac{\log(1 + \theta\lambda)}{\theta} d\theta \right) \\ &\leq O_\delta(\log \theta_0^{-1} + \Delta\lambda). \end{aligned}$$

The fugacity condition gives $\Delta\lambda \leq (1 - \delta)x$ and $\log \theta_0^{-1} = \log(2\Delta\lambda) = O(\log(2 + x))$. Moreover, $a \leq \|A_G\|_2 \leq \Delta$, so $x = \Delta/(a - 1) > 1$ and $\log(2 + x) = O(x)$. Therefore $B_F = O_\delta(x)$, and [theorem 2.5](#) gives

$$C_{\text{AT}} \leq C_0 e^{B_F} = \exp(O_\delta(x)), \quad \alpha_{\text{Ent}}(P_\mu^{\text{GD}}) \geq n^{-1} \exp(-O_\delta(x)).$$

By Fact 3.5 in [4], if ATE for measure μ holds with constant C_{AT} then the mixing time of Glauber dynamics is $C_{\text{AT}}n \log \log(1/\mu_{\min})$. Since $\mu_{\min} \geq \left(\frac{\min\{1, \lambda\}}{1+\lambda}\right)^n$ in our case, this proves the mixing bound. $\square$

Proof of Corollary 1.2. For fixed $\Delta \geq 3$, the extreme-eigenvalue estimates in [15] give

$$a = -\lambda_{\min}(A_G) = (2 + o_n(1))\sqrt{\Delta - 1}$$

with probability $1 - o_n(1)$. On this event, $a > 1$ and $a = \Theta(\sqrt{\Delta})$, and the assumed fugacity bound implies $\lambda(a - 1) \leq 1 - \frac{\delta}{2}$ for all sufficiently large n . Apply theorem 1.1 with $\frac{\delta}{2}$. $\square$

Remaining question. It remains open whether the exponential degree dependence can be replaced by a polynomial, giving $O_\delta(\text{poly}(\Delta)n \log n)$ mixing throughout the same fugacity regime.

A Hard-core easy-regime approximate tensorization

The proof uses the following three-state inequality.

Lemma A.1 (Three-state logarithmic inequality). *Let $0 < p \leq 1/2$, $q = 1 - p$, and $x, y > 0$. Then*

$$\begin{aligned} E_p(x, y) &= (x - 1) \log(q + py) + (y - 1) \log(q + px) \\ &\quad + p(x - 1) \log x + p(y - 1) \log y \geq 0. \end{aligned} \quad (11)$$

Proof. If $(x - 1)(y - 1) \geq 0$, then the signs of the first two logarithms are determined by $q + py - 1 = p(y - 1)$ and $q + px - 1 = p(x - 1)$; the last two terms are nonnegative because $(t - 1) \log t \geq 0$. Thus every term is nonnegative.

By symmetry it remains to consider opposite signs. Write

$$x = 1 + u > 1, \quad y = 1 - v < 1, \quad u > 0, \quad 0 < v < 1,$$

and define

$$h(t) = \begin{cases} \log(1 + t)/t, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

A direct rearrangement gives

$$E_p(x, y) = p\Phi_p, \quad \Phi_p = u^2 h(u) + v^2 h(-v) - uv(h(pu) + h(-pv)). \quad (12)$$

For $t > -1$,

$$h(t) = \int_0^1 \frac{ds}{1 + st}, \quad h''(t) = 2 \int_0^1 \frac{s^2 ds}{(1 + st)^3} > 0.$$

Hence

$$\Phi_p'' = -uv(u^2 h''(pu) + v^2 h''(-pv)) \leq 0.$$

Thus $p \mapsto \Phi_p$ is concave on $[0, 1/2]$ and

$$\Phi_p \geq (1 - 2p)\Phi_0 + 2p\Phi_{1/2}. \quad (13)$$

Set $r = v/u > 0$, $t = v \in (0, 1)$, so $u = t/r$. At $p = 0$,

$$\Phi_0 = uG_r(t), \quad G_r(t) = \log\left(1 + \frac{t}{r}\right) - r \log(1 - t) - 2t.$$

Differentiation gives

$$G'_r(t) = \frac{(r+t-1)(r+2t-1)}{(r+t)(1-t)}.$$

If $r \geq 1$, G_r is increasing. If $0 < r < 1$, its minimum can occur only at $t = 0$ or $t = 1 - r$. Now

$$G_r(0) = 0, \quad G_r(1-r) = -(1+r) \log r - 2(1-r) \geq 0,$$

where the last inequality follows from

$$-\log r \geq \frac{2(1-r)}{1+r}, \quad 0 < r \leq 1.$$

Therefore $\Phi_0 \geq 0$.

At $p = 1/2$,

$$\Phi_{1/2} = uH_r(t),$$

where

$$H_r(t) = \log \left(1 + \frac{t}{r}\right) + 2 \log \left(1 - \frac{t}{2}\right) - r \log(1-t) - 2r \log \left(1 + \frac{t}{2r}\right).$$

A direct differentiation gives

$$H'_r(t) = \frac{(r+t-1)(2r+3t-2)(2r+t-rt)}{(r+t)(2r+t)(2-t)(1-t)}.$$

The last numerator factor is positive. For $r \geq 1$, H_r is increasing; for $0 < r < 1$, the same monotonicity analysis shows that the minimum occurs only at $t = 0$ or $t = 1 - r$, and

$$H_r(0) = 0, \quad H_r(1-r) = (r-1) \log \frac{4r}{(1+r)^2} \geq 0,$$

because $4r \leq (1+r)^2$. Hence $\Phi_{1/2} \geq 0$. The chord bound (13) proves (11). $\square$

Proof of Theorem 3.3. Set $p = \rho/(1+\rho)$ and $q = 1 - p$. Let $P_i = K_i^{\mu_\rho}$ and consider the continuous-time heat-bath generator

$$L = \sum_{i \in V} (P_i - I).$$

For $f > 0$, define

$$D_i(f) = \mu_\rho[(f - P_i f) \log f], \quad D(f) = \sum_i D_i(f),$$

and

$$Q_{ik}(f) = \mu_\rho \left[(f - P_i f) \log \frac{f}{P_k f} \right].$$

Since $\log P_i f$ is measurable off i ,

$$D_i(f) = \mu_\rho[P_i(f \log f) - P_i f P_i \log f] \geq 0.$$

We first prove the cross-term estimate

$$\sum_{i,k \in V} Q_{ik}(f) \geq (1 - \Delta p) D(f). \tag{14}$$

For an edge $\{i, k\} \in E$, the required local inequality is

$$Q_{ik}(f) + Q_{ki}(f) \geq -p(D_i(f) + D_k(f)). \quad (15)$$

Condition on all coordinates outside $\{i, k\}$. If either endpoint is blocked by an occupied external neighbor, the corresponding update is deterministic and the assertion follows from nonnegativity of D_i, D_k . Otherwise the allowed states are 00, 10, 01, with weights proportional to $1, \rho, \rho$. Write

$$a = f(00), \quad b = f(10), \quad c = f(01), \quad N = 1 + 2\rho,$$

and

$$u = qa + pb, \quad v = qa + pc.$$

Writing D_i, D_k, Q_{ik}, Q_{ki} temporarily for their conditional contributions, a direct calculation gives

$$\begin{aligned} D_i(f) &= \frac{p}{N}(b - a)(\log b - \log a), & D_k(f) &= \frac{p}{N}(c - a)(\log c - \log a), \\ Q_{ik}(f) &= \frac{p}{N}(b - a) \log \frac{v}{a}, & Q_{ki}(f) &= \frac{p}{N}(c - a) \log \frac{u}{a}. \end{aligned}$$

After setting $x = b/a$ and $y = c/a$, the desired conditional inequality is exactly (11); averaging over the outside configuration proves (15). The theorem assumes $\rho \leq 1/\Delta \leq 1$, so $p \leq 1/2$.

For $i = k$,

$$Q_{ii}(f) = D_i(f),$$

because the term containing $\log P_i f$ has zero expectation against $f - P_i f$.

If $i \neq k$ and $\{i, k\} \notin E$, then, conditional on all other coordinates, i and k are independent, possibly with deterministic coordinates. Hence $P_i P_k = P_k P_i$ and

$$Q_{ik}(f) = D_i(f) - D_i(P_k f).$$

The function

$$J(a, b) = (a - b)(\log a - \log b)$$

is jointly convex on $(0, \infty)^2$, since

$$\nabla^2 J(a, b) = \begin{pmatrix} (a+b)/a^2 & -(a+b)/(ab) \\ -(a+b)/(ab) & (a+b)/b^2 \end{pmatrix} \succeq 0.$$

Writing D_i as one half of the conditional average of J over two independent copies of coordinate i , Jensen's inequality in coordinate k gives $D_i(P_k f) \leq D_i(f)$. Thus $Q_{ik}(f) \geq 0$ for nonadjacent sites.

Pair the two orientations of every edge and apply (15). We obtain

$$\begin{aligned} \sum_{i,k} Q_{ik}(f) &\geq \sum_i D_i(f) - p \sum_{\{i,k\} \in E} (D_i(f) + D_k(f)) \\ &= D(f) - p \sum_i d_i D_i(f) \\ &\geq (1 - \Delta p) D(f). \end{aligned}$$

Let $f_t = e^{tL} f$ for $f > 0$. The entropy-dissipation identity is

$$-\frac{d}{dt} \text{Ent}_{\mu_\rho}(f_t) = D(f_t).$$

The finite heat-bath chain is irreducible and $f_t \rightarrow \mu_\rho(f)$, so

$$\text{Ent}_{\mu_\rho}(f) = \int_0^\infty D(f_t) dt. \quad (16)$$

For a fixed k , set

$$\Psi_k(t) = \mu_\rho[\text{Ent}_k(f_t)].$$

Using self-adjointness of P_k in $L^2(\mu_\rho)$,

$$\Psi'_k(t) = \mu_\rho \left[L f_t \log \frac{f_t}{P_k f_t} \right] = - \sum_i Q_{ik}(f_t).$$

Since $\Psi_k(t) \rightarrow 0$,

$$\mu_\rho[\text{Ent}_k(f)] = \int_0^\infty \sum_i Q_{ik}(f_t) dt.$$

Sum over k , apply (14), and use (16):

$$\sum_k \mu_\rho[\text{Ent}_k(f)] \geq (1 - \Delta p) \text{Ent}_{\mu_\rho}(f).$$

Finally,

$$1 - \Delta p = \frac{1 - \rho(\Delta - 1)}{1 + \rho}.$$

For a nonnegative f , apply the positive result to $f + \varepsilon$ and let $\varepsilon \downarrow 0$. $\square$

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